A Comparative Study of Graph Neural Networks and Traditional Machine Learning for Molecular Toxicity Prediction

Mandana Roosta-Reza Jafari

Physics Department of Shahid Beheshti University

The prediction of molecular toxicity is a critical challenge in pharmaceutical development, where early identification of harmful compounds can save significant time and resources. While Graph Neural Networks (GNNs) have shown promise for molecular property prediction by leveraging graph structure, their effectiveness compared to traditional machine learning methods on small datasets remains unclear. This study compares the performance of a Graph Convolutional Network against traditional machine learning baselines (Random Forest and Logistic Regression) for predicting molecular mutagenicity using the MUTAG dataset. Our results demonstrate that traditional feature-based methods significantly outperform the GNN approach (86.8% vs 65.8% accuracy), highlighting the importance of dataset size and feature engineering in method selection for molecular property prediction tasks.

Keywords: Graph Neural Networks, Molecular Property Prediction, Machine Learning, Toxicity Prediction, Comparative Study

---

1. Introduction

The identification of toxic molecular compounds is a fundamental challenge in pharmaceutical research and drug development. Traditional approaches rely on extensive laboratory testing, which is both time-consuming and expensive. Computational methods for toxicity prediction offer the potential to screen large numbers of compounds rapidly, reducing the need for animal testing and accelerating drug discovery pipelines.

Recent advances in Graph Neural Networks (GNNs) have generated significant interest in molecular property prediction tasks. GNNs can naturally represent molecules as graphs where atoms serve as nodes and chemical bonds as edges, potentially capturing structural relationships that traditional feature-based methods might miss. However, the relative performance of GNNs compared to well-established machine learning approaches on small datasets remains an open question.

This study addresses the following research question: *Can Graph Neural Networks outperform traditional machine learning methods for molecular toxicity prediction when working with small datasets?* We conduct a systematic comparison using the MUTAG dataset, comparing a Graph

Convolutional Network implementation against Random Forest and Logistic Regression baselines.

2. Related Work

2.1 Molecular Property Prediction

Computational prediction of molecular properties has been an active area of research for decades. Traditional approaches have relied on molecular descriptors—numerical features that encode various aspects of molecular structure and properties. These descriptors can be categorized into constitutional, topological, geometric, and electronic descriptors.

2.2 Graph Neural Networks for Molecules

Graph Neural Networks have emerged as a powerful paradigm for learning from graph-structured data. For molecular applications, several architectures have been proposed, including Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), and Message Passing Neural Networks (MPNNs). These methods have shown promising results on various molecular property prediction benchmarks.

2.3 Small Dataset Challenges

A persistent challenge in molecular machine learning is the availability of high-quality labeled data. Many molecular property datasets contain only hundreds or thousands of examples, which may be insufficient for training complex deep learning models effectively. This limitation motivates our comparative study on a small dataset.

3. Methodology

3.1 Dataset

We used the MUTAG dataset, a standard benchmark for molecular classification containing 188 nitroaromatic compounds labeled as mutagenic or non-mutagenic. The dataset characteristics are:

- Total molecules: 188
- Classes: 2 (mutagenic/non-mutagenic)
- Average molecular size: 17.9 atoms
- Node features: 7 per atom
- Train/test split: 150/38 (80%/20%)

3.2 Graph Neural Network Architecture

We implemented a Graph Convolutional Network with the following architecture:

- 3 Graph Convolutional layers (64 hidden units each)
- ReLU activation functions
- Dropout regularization (p=0.2)
- Global mean pooling for graph-level representation
- Linear classifier for binary classification
- Total parameters: 8,962

The model was trained using the Adam optimizer with a learning rate of 0.01 and cross-entropy loss for 50 epochs.

3.3 Baseline Methods

For traditional machine learning baselines, we extracted molecular features including:

- Basic graph statistics (number of atoms, bonds)
- Node feature statistics (mean and standard deviation)
- Total feature dimensionality: 16

We compared against:

1. Random Forest: 100 estimators with default parameters
2. Logistic Regression: L2 regularization with maximum 1000 iterations
3. Random Baseline: Majority class prediction

3.4 Evaluation Metrics

We evaluated all methods using:

- Classification accuracy
- Precision, recall, and F1-score for each class
- Confusion matrices for detailed error analysis

4. Results

4.1 Overall Performance Comparison

Table 1 presents the test accuracy results for all methods:

Method	Test Accuracy	Standard Error
Logistic Regression	86.8%	±5.4%
Random Forest	84.2%	±5.8%
Graph Neural Network	65.8%	±7.7%
Random Baseline	55.3%	±8.0%

Traditional machine learning methods significantly outperformed the GNN approach by over 20 percentage points.

4.2 Detailed Classification Performance

The detailed classification report for the GNN shows:

Class	Precision	Recall	F1-Score	Support
Safe	0.75	0.35	0.48	17
Mutagenic	0.63	0.90	0.75	21

The model exhibited better performance on the mutagenic class, likely due to class imbalance in the training set (69% mutagenic molecules).

4.3 Training Dynamics

The GNN showed signs of overfitting:

- Final training accuracy: 77.3%
- Final test accuracy: 65.8%
- Gap indicates poor generalization to unseen data

Training completed in 1.87 seconds, demonstrating computational efficiency.

5. Discussion

5.1 Why Traditional Methods Succeeded

The superior performance of traditional machine learning methods can be attributed to several factors:

1. Dataset Size: With only 150 training examples, the dataset is too small to effectively train deep neural networks, which typically require thousands of examples.
2. Feature Engineering: The handcrafted molecular descriptors captured essential chemical patterns more efficiently than learning from raw graph structure.
3. Model Complexity: Simpler models with fewer parameters were better suited to the available data, avoiding overfitting.
4. Domain Knowledge: Traditional descriptors incorporated decades of chemical knowledge, providing a strong inductive bias.

5.2 GNN Limitations on Small Data

The Graph Neural Network's underperformance reveals important limitations:

1. Data Hunger: Deep learning models require large datasets to learn effective representations from scratch.
2. Overfitting: The 12 percentage point gap between training and test accuracy indicates the model memorized training data rather than learning generalizable patterns.
3. Limited Inductive Bias: Without sufficient data, the GNN could not discover the chemical patterns that traditional descriptors encode explicitly.

5.3 Implications for Practice

These results have important implications for practitioners:

1. Method Selection: Dataset size should be a primary consideration when choosing between traditional ML and deep learning approaches.
2. Feature Engineering Value: Domain expertise remains highly valuable, especially for small datasets.
3. Baseline Importance: Simple methods should always be included as baselines to validate the need for complex approaches.

6. Limitations and Future Work

6.1 Study Limitations

1. Single Dataset: Results are based on one small dataset and may not generalize to other molecular property prediction tasks.
2. Simple GNN Architecture: More sophisticated graph neural network architectures might perform better.
3. Limited Hyperparameter Tuning: Both traditional methods and the GNN used default or minimally tuned parameters.
4. Feature Engineering: Traditional methods benefited from manual feature engineering while the GNN learned from raw structure.

6.2 Future Research Directions

1. Larger Datasets: Investigate performance differences on datasets with thousands of molecules.
2. Hybrid Approaches: Combine graph neural networks with traditional molecular descriptors.
3. Architecture Exploration: Test more sophisticated GNN variants (Graph Attention Networks, Graph Transformers).
4. Transfer Learning: Pre-train GNNs on large molecular datasets and fine-tune on small target datasets.

7. Conclusion

This comparative study demonstrates that traditional machine learning methods significantly outperform Graph Neural Networks for molecular toxicity prediction on small datasets. Logistic Regression achieved 86.8% accuracy compared to 65.8% for the GNN approach, highlighting the continued relevance of feature-based methods when data is limited.

The results emphasize several key principles for molecular machine learning:

1. Dataset size critically affects method selection - small datasets favor traditional ML approaches
2. Feature engineering remains valuable when domain knowledge exists and data is limited
3. Model complexity must match available data to avoid overfitting
4. Proper baselines are essential for evaluating the need for complex methods

While Graph Neural Networks show promise for molecular property prediction, practitioners working with small datasets should first establish strong traditional machine learning baselines. The decision to employ more complex graph-based methods should be justified by demonstrated performance improvements over simpler alternatives.

This work contributes to the growing understanding of when and how to apply different machine learning paradigms in computational chemistry and drug discovery.

References

1. Duvenaud, D. K., et al. (2015). Convolutional networks on graphs for learning molecular fingerprints. *Advances in Neural Information Processing Systems*.
2. Gilmer, J., et al. (2017). Neural message passing for quantum chemistry. *International Conference on Machine Learning*.
3. Kipf, T. N., & Welling, M. (2016). Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations*.
4. Morris, C., et al. (2020). TUDataset: A collection of benchmark datasets for learning with graphs. *ICML 2020 Workshop on Graph Representation Learning*.
5. Delaney, J. S. (2004). ESOL: estimating aqueous solubility directly from molecular structure. *Journal of Chemical Information and Computer Sciences*.
6. Rogers, D., & Hahn, M. (2010). Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*.
7. Wu, Z., et al. (2018). MoleculeNet: a benchmark for molecular machine learning. *Chemical Science*.
8. Yang, K., et al. (2019). Analyzing learned molecular representations for property prediction. *Journal of Chemical Information and Modeling*.

Appendix A: Implementation Details

A.1 Software Environment

- Python 3.11.13
- PyTorch Geometric 2.6.1
- Scikit-learn 1.3.0
- RDKit 2025.3.6
- CUDA acceleration enabled

A.2 Hyperparameters

Graph Neural Network:

- Learning rate: 0.01
- Batch size: 32
- Hidden dimensions: 64
- Dropout rate: 0.2
- Training epochs: 50

Random Forest:

- Number of estimators: 100
- Random state: 42
- Default scikit-learn parameters

Logistic Regression:

- Maximum iterations: 1000
- Random state: 42
- L2 regularization (default)

A.3 Computational Resources

- Training time: < 2 seconds
- Hardware: Kaggle GPU environment
- Memory usage: < 1GB

Appendix B: Additional Results

B.1 Learning Curves

The GNN showed overfitting behavior with training accuracy consistently higher than validation accuracy after epoch 10.

B.2 Feature Importance

For Random Forest, the most important features were:

1. Number of atoms (0.23)
2. Number of bonds (0.19)
3. Mean node feature values (0.58 combined)